Evan Yarnall

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RESEARCH INTERESTS

Sensorimotor development and integration in infants and children with or at risk for cerebral palsy; development of bimanual skills and their cortical correlates; corticospinal connectivity and online motor control, using fNIRS, TMS and behavioral coding.

EDUCATION

PhD, Rehabilitation Science · Boston University, Sargent College, Boston, MA — 2023 – 2028 (expected)
Development, Experience, and Motor Recovery (DEMR) Lab. Advisor: Claudio L. Ferre, PhD. Dissertation: a longitudinal study of sensorimotor development in infants at risk for cerebral palsy.

BS, Kinesiology (Pre-Physical Therapy) · Temple University, College of Public Health, Philadelphia, PA — 2015 – 2019

RESEARCH EXPERIENCE

Graduate Researcher, DEMR Lab · Boston University, Sargent College — 2023 – present

- Study sensorimotor integration in children with unilateral cerebral palsy, combining fNIRS and TMS to assess corticospinal connectivity and sensory cortical responses during reaching.
- Analyze (R, MATLAB, Python) a longitudinal sensorimotor dataset following typically developing infants and infants at risk for cerebral palsy from 6 to 24 months (dissertation dataset).
- Designed a moving-target reaching apparatus for testing online motor control, built with lead engineer Julia Zayas (BU College of Engineering).
- Develop and maintain behavioral coding pipelines in Datavyu (Ruby, Python), including inter-rater reliability workflows.

Research Technician, Laboratory for Human Neurobiology · Boston University School of Medicine — 2022 – 2023

- Used wearable technology and behavioral coding to measure behavior in children with atopic dermatitis.
- Led outreach to local healthcare facilities to recruit study participants.

Research Technician (part-time) · Symmetry Science Consulting — 2022 – 2023

- Processed wearable device data (Apple Watch, actigraphy) and participant-reported sleep and skin quality surveys.

Clinical Research Coordinator · Shriners Hospitals for Children, Philadelphia, PA — 2021 – 2022

- Managed industry-funded clinical trials on adolescent orthopedic conditions and spinal cord injury.
- Prepared IRB submissions for FDA-regulated studies and contributed to grant writing.

Research Assistant, Virtual Environment and Postural Orientation (VEPO) Lab · Temple University — 2018

- Supported EEG data collection and analysis and processed motion capture data for research on postural orientation.

Research Assistant, Temple Research Immersive Balance and Locomotion (TRIBAL) Lab · Temple University — 2016 – 2017

- Set up motion capture (Vicon) markers and cameras and performed preliminary analysis of movement data.
- Supported participant-facing study procedures, including explaining experimental protocols.

PUBLICATIONS

Underline: Evan Yarnall.

PEER-REVIEWED

P1 Brenn BR, DiSilvio GM, **Yarnall E**, Steindler J, Tarazi S, Rompala A, Akhnoukh K, Choudhry DK. (2024). A Comparative Evaluation of Unilateral and Bilateral Sequential Lung Isolation for Vertebral Body Tethering: A Retrospective Propensity Matched Analysis. *Cureus*, 16(5), e59723. doi:10.7759/cureus.59723

UNDER REVIEW

P2 Su X, Kim H, Kelly M, **Yarnall E**, Ferre CL. Comparative Effectiveness of Early Therapies for Infants and Young Children with or at-risk for Cerebral Palsy: A Network Meta-analysis. *Neurorehabilitation and Neural Repair*.

P3 Su X, Kim H, Kelly M, **Yarnall E**, Ferre CL. fNIRS Commons: An Open Dataset for Studying Cortical Activation during Spontaneous Movement in Infancy. *Neurophotonics*.

IN PREPARATION

P4 **Yarnall E**, Su X, Kelly M, Ferre CL. Lateralized Cortical Activity during the Development of Role-Differentiated Bimanual Manipulation.

P5 Ferre CL, Su X, Kim H, **Yarnall E**, Kelly M. Spontaneous Movements and Development of Cortical Function: An fNIRS Study.

P6 Su X, Kelly M, **Yarnall E**, Ferre CL. Comparison of Spontaneous Movement Measurements Using Video Observation and Inertial Measurement Units (IMUs) in Typically Developing Infants and Infants at Risk for Cerebral Palsy.

P7 Kim H, Su X, Kelly M, **Yarnall E**, Ferre CL. Characterizing Upper-Limb Force Control and Its Underlying Neurophysiological Correlates in Neurotypical Adults.

CONFERENCE PRESENTATIONS

AACPDM: American Academy for Cerebral Palsy and Developmental Medicine. ASNR: American Society of Neurorehabilitation.

C1 **Yarnall E**, Kelly M, Kim H, Su X, Ferre CL. (2026). Longitudinal Trajectories of Bilateral Hand Use: Changes in Behavior and Cortical Activity. *AACPDM 80th Annual Meeting*, Philadelphia, PA. `Scientific poster SP20`

C2 **Yarnall E**, Zayas J, Ferre CL. (2026). Moving Target Board for Testing Online Motor Control. *AACPDM 80th Annual Meeting*, Philadelphia, PA. `Demonstration poster DP26`

C3 Kim H, **Yarnall E**, Ferre CL. (2026). Moderate Intensity Upper Limb Resistance Band Training Improves Neurophysiology in a Young Adult With Spastic Diplegic Cerebral Palsy: A Single Case Design Study. *AACPDM 80th Annual Meeting*, Philadelphia, PA. `Clinical observation poster CO18`

C4 Su X, Kelly M, Kim H, **Yarnall E**, Ferre CL. (2026). Feasibility of Using DeepLabCut for Markerless Motion Tracking During Infant Reaching. *AACPDM 80th Annual Meeting*, Philadelphia, PA. `Scientific poster SP167`

C5 **Yarnall E**, Su X, Kim H, Kelly M, Ferre CL. (2025). Motor Experience Drives Brain Development: A Longitudinal Study of Infant Manual Skills and Cortical Activity in Neurotypical Infants and Infants with Unilateral Cerebral Palsy. *AACPDM 79th Annual Meeting*, New Orleans, LA. `Free paper R1`

C6 Su X, **Yarnall E**, Kelly M, Kim H, Ferre CL. (2025). Comparison of Spontaneous Movement Measurements Using Video Observation and Inertial Measurement Units (IMUs) in Typically Developing Infants and Infants at Risk for Cerebral Palsy. *AACPDM 79th Annual Meeting*, New Orleans, LA. `Free paper Q2`

C7 Su X, Kelly M, Kim H, **Yarnall E**, Ferre CL. (2024). Exploring the Dynamics of Spontaneous Infant Movement: A Longitudinal Study of Movement Patterns and Cortical Activity. *AACPDM 78th Annual Meeting*, Quebec City, Canada. `Scientific poster SP169`

C8 Su X, Kim H, **Yarnall E**, Kelly M, Ferre CL. (2024). Spontaneous Movements and Cortical Activity during Early Infancy. *ASNR Annual Meeting*, San Antonio, TX. `Poster P120`

AWARDS

Student Scholarship · American Academy for Cerebral Palsy and Developmental Medicine 2025

OUTREACH AND VOLUNTEERING

Volunteer · CP Soccer New England 2024 – present
Soccer programs for kids with cerebral palsy and other neurological conditions.

Community outreach · DEMR Lab, Boston University 2024 – present
Hands-on brain and movement activities for families at Science on State Street (Framingham State University), Open Streets Boston (Allston/Brighton, Roxbury), and Play to Thrive.

SKILLS

Neuroimaging & neurophysiology	fNIRS, TMS, EEG
Movement & behavior	Motion capture (Vicon), wearable technology, behavioral video coding (Datavyu)
Analysis & programming	Statistical data analysis (linear mixed-effects models), inter-rater reliability; R, Python, MATLAB, Ruby
Clinical research	Human subjects research, clinical trials, Good Clinical Practice (GCP), IRB submissions for FDA-regulated studies, regulatory compliance, participant recruitment and support, grant writing

CERTIFICATIONS

RCR Foundations (Responsible Conduct of Research) · CITI Program 2024

Biomedical Research · CITI Program 2022

Good Clinical Practice (GCP) · CITI Program 2021